Train Reservation System

Using C++ & Object-Oriented Programming

Developed by: **Harshith Nellore** | Mini Project Presentation

Introduction & Project Overview

What Is It?

A console-based C++ application that simulates a real-world train ticket booking experience — search, reserve, and generate tickets digitally.

Purpose

Demonstrate core OOP principles through a practical, relatable project that mirrors industry-level software design thinking.

Why Automation?

Manual booking is slow and error-prone. Automation improves speed, accuracy, and scalability — key pillars of modern software systems.

Objectives & Features

Ticket Reservation

Book seats on available trains with passenger details stored securely.

Train Search

Look up trains by route, number, or destination in real time.

Seat Availability

Displays live seat counts and prevents double booking automatically.

Reservation Storage

All bookings are persisted using file handling for data continuity.

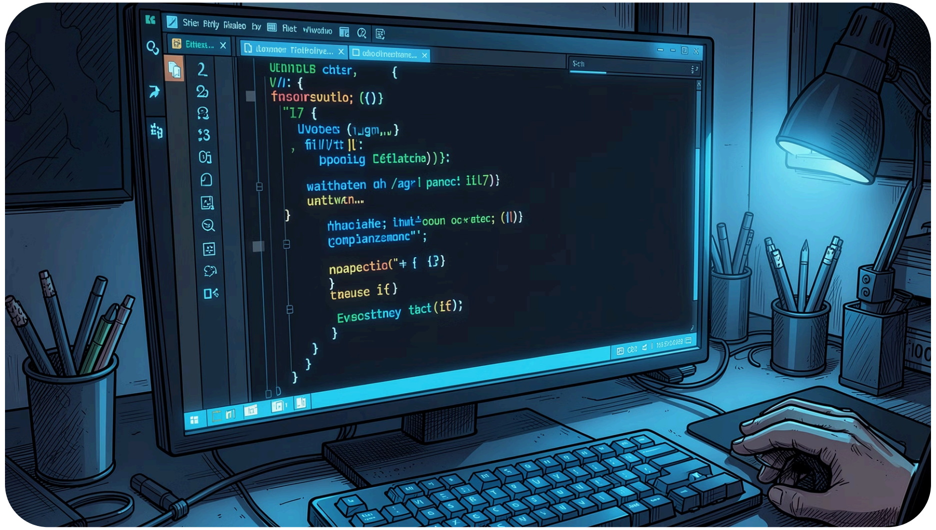
Ticket Generation

Formatted ticket output with passenger name, train, and seat info.

File Handling

Data is read and written using C++ file streams for persistence.

Technologies & OOP Concepts Used



Language: C++

Industry-standard, compiled, high-performance systems language.

OOP Pillars

Classes, Objects, Encapsulation, and Constructors drive the design.

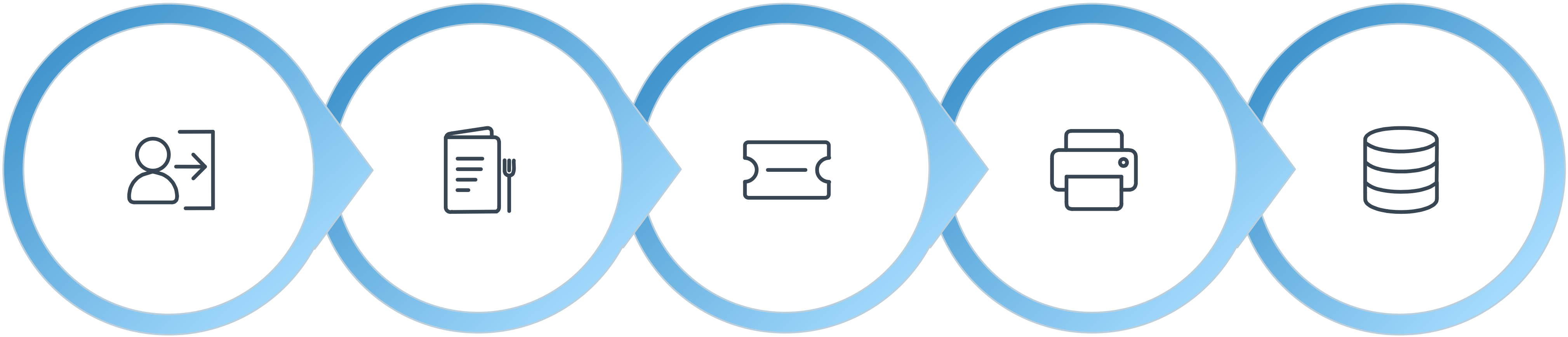
STL Vector

Dynamic storage for passenger and train records at runtime.

File Handling

`fstream` used for reading/writing reservation data to disk.

System Architecture & Workflow



The system follows a linear, menu-driven flow. After authentication, users navigate the main menu to search trains, make reservations, and receive a printed ticket — all records are automatically saved to file storage for future retrieval.

Classes & Code Snippets

Passenger

Stores name, age, seat number, and booking ID.

Train

Holds train number, route, total seats, and availability.

TrainReservationSystem

Orchestrates booking logic, search, and file I/O operations.

```
// Passenger Constructor
Passenger::Passenger(string n, int a, int seat) {
 name = n; age = a;
 seatNo = seat;
}

// Reserve Function
void TrainReservationSystem::reserve(
 string trainNo, Passenger p) {
 Train* t = findTrain(trainNo);
 if(t->availableSeats > 0) {
 t->availableSeats--;
 passengers.push_back(p);
 saveToFile(p);
 }
}
```

Advantages & Future Enhancements

CURRENT STRENGTHS

Beginner Friendly

Clear class structure ideal for learning OOP from the ground up.

Real-World Logic

Mirrors actual booking workflows, making it portfolio-worthy.

OOP + File Handling

Encapsulation and persistence demonstrated together in one project.

ROADMAP

01

GUI Version

Upgrade to a graphical interface using Qt or a web front-end.

02

Database Integration

Replace file storage with SQL/MySQL for scalable data management.

03

Online Payment & PNR

Add payment gateway simulation and auto-generated PNR numbers.

04

User Registration

Multi-user login system with authentication and booking history.

Conclusion & Thank You

Project Summary

A fully functional console-based Train Reservation System built in C++, demonstrating OOP, STL, and file handling in a real-world context.

Why OOP Matters

OOP principles — encapsulation, modularity, and reusability — are the foundation of scalable, maintainable modern software engineering.

Questions & Answers

Thank you for your time and attention. Open to all questions from the panel — happy to walk through any part of the code or design.

Developed by Harshith Nellore | Mini Project — C++ & OOP